InteriorVision: Classification and Generation of Interior Design Images

Retaj Alhazmi (445006594) **Maria Mamdooh** (445006784)
Ghala Shahir (445006326) **Ghadeer Sami** (44511798)
Rawan Qader (445013919)
Umm Al-Qura University

Department of Artificial Intelligence

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Abstract

This project presents a dual-track deep learning framework for interior design analysis and generation. The system integrates two complementary models trained on a curated dataset of 4,000 images: a discriminative track employing an enhanced multi-task ResNet50 for image classification, and a generative track utilizing Stable Diffusion with Low-Rank Adaptation (LoRA) for text-to-image synthesis.
To improve classification, the baseline CNN was optimized via full backbone fine-tuning, dropout regularization, batch normalization, and label smoothing. The enhanced CNN achieved 74% room-type accuracy and 60% style accuracy, significantly outperforming the baseline. For the generative task, the LoRA-augmented Stable Diffusion model successfully synthesized domain-specific interior concepts from text prompts. While qualitative results demonstrated high stylistic consistency, quantitative evaluation via CLIP scores highlighted the strict dependencies of diffusion models on dataset scale and hardware constraints. Overall, this framework demonstrates the efficacy of combining discriminative and generative deep learning pipelines for specialized design domains under restricted data and hardware regimes.

1 Introduction

1.1 Problem Statement

Interior design images exhibit high variability in lighting, textures, spatial layout, and stylistic elements. Traditional computer vision techniques struggle to capture these abstract and multi-level patterns, which limits their ability to classify or generate realistic interior scenes. The challenge lies in learning both local visual cues (e.g., edges, materials, furniture shapes) and global semantic structure (e.g., room layout, style coherence). Deep learning provides a scalable solution by automatically learning hierarchical representations that can generalize across diverse interior environments.

1.2 Why Deep Learning is Suitable

Deep learning models particularly Convolutional Neural Networks (CNNs) and Latent Diffusion Models are well-suited for this domain because they:

- Extract multi-level spatial features directly from raw images.
- Capture global context such as style, color harmony, and room composition.
- Learn complex mappings between text prompts and visual outputs.
- Adapt efficiently to new data distributions through fine-tuning.

This makes deep learning ideal for both discriminative tasks (classification) and generative tasks (image synthesis).

1.3 Objectives

The main goal of this project is to build, analyze, and improve two different computational tracks to handle our image dataset. The specific objectives are divided into two main parts:

The Discriminative Track (Image Classification)

- Implement ResNet50 as the baseline classification model.
- Analyze how bottleneck blocks and residual skip connections extract deep features without vanishing gradients.
- Evaluate classification performance using Global Average Pooling (GAP) and Softmax.

The Generative Track (Image Synthesis)

- Deploy Stable Diffusion as the baseline for text-to-image generation.
- Study how a Variational Autoencoder (VAE) compresses images into an efficient latent space.

Optimization & Enhancements

- Propose and test structural and algorithmic improvements for both baseline models.
- Enhance training speed (convergence), generalizability, and overall performance metrics.

2 Dataset

2.1 Dataset Description

This project uses the *Interior Design Images & Metadata Dataset* from Kaggle. The dataset contains interior design images grouped by room type and design style. It was chosen because it provides both images and structured labels, which makes it suitable for the classification and image generation parts of the project.

The dataset includes 4,139 images from four room categories: bathroom, bedroom, kitchen, and living room. It also includes five design styles: boho, industrial, minimalist, modern, and scandinavian. Each image has metadata such as the image path, room type, style, and encoded label. The dataset is already divided into training, validation, and testing sets, which helps us train and evaluate the models in an organized way.

In this project, the dataset is used for two main tasks. First, it is used to train CNN models to classify the room type and interior style. Second, the room type and style labels are used to create text prompts for the image generation model. These prompts help the Stable Diffusion model generate interior design images that match the given description.

The operational volume of the dataset was optimized through a targeted data cleaning process. A detailed breakdown of the distribution splits before and after this cleaning stage is compiled in Table 1.

Table 1: Summary of dataset splits before and after data cleaning.

Split	Before	After
Training	2,648	2,556
Validation	663	639
Testing	828	805
Total	**4,139**	**4,000**

2.2 Data Preprocessing

Before training the models, several preprocessing steps were applied to prepare the dataset and make it suitable for both classification and image generation tasks.

2.2.1 Path Correction and Cleaning

The image paths provided in the CSV files were checked and corrected to match the working directory used in Google Colab. Images with invalid or missing paths were removed when necessary to avoid loading errors during training.

2.2.2 Prompt Generation

For the generative part of the project, text prompts were automatically created from the metadata by combining the design style and room type. For example, an image labeled as *modern* and *kitchen* was converted into the prompt: `"modern kitchen interior"`. These prompts were used to guide the Stable Diffusion model during image generation and LoRA fine-tuning.

2.2.3 Image Resizing and Format Conversion

All images were converted to RGB format and resized to a fixed size before being passed to the models. The classification model used resized images suitable for the CNN input, while the generative model used images prepared for Stable Diffusion training.

2.2.4 Normalization and Augmentation

For the CNN classifier, image pixel values were normalized before training. In the improved model, data augmentation was applied to the training images to increase diversity and reduce overfitting. The augmentation included transformations such as horizontal flipping, rotation, cropping, and color adjustment.

2.2.5 Label Encoding

The room type and style labels were converted into numerical labels so they could be used by the CNN classification model. For the diffusion model, label encoding was not required because the generated text prompts were used as conditioning inputs.

2.3 Sample Images

To inspect the visual diversity of the data, sample representations were extracted directly from the training pipelines. Figure 1 demonstrates sample images from the training dataset covering different room types and interior design styles, alongside their associated prompt strings.

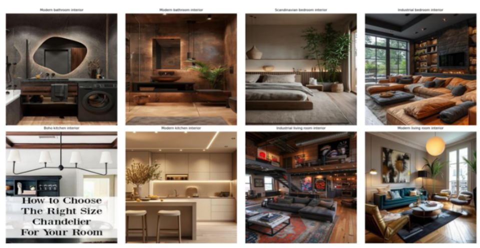

Figure 1: Sample images from the training dataset showing different styles.

3 Methodology

This section delineates the architectural design, layer configurations, and optimization frameworks utilized across the two distinct computational tracks of this project:

- **The Discriminative Track**: Utilizing Deep CNNs for image classification.
- **The Generative Track**: Utilizing Latent Diffusion Models for conditional image synthesis.

  Each section analyzes the baseline configurations followed by the proposed structural and algorithmic enhancements designed to optimize convergence, generalizability, and performance.

3.1 Baseline CNN (ResNet50): Layers and Structure

The discriminative baseline model is anchored on the standard ResNet50 architecture, a 50-layer deep convolutional network engineered to overcome the vanishing gradient problem through the integration of residual skip connections [1].

3.1.1 Layer-by-Layer Macro-Structure

- **Stage 0:** The network accepts an input image volume shaped $224 \times 224 \times 3$. It processes it through an initial 7×7 convolutional layer featuring 64 filters, applied with a stride of 2 and padding of 3. This is followed by BN and ReLU. Spatial dimensions are halved via a 3×3 Max Pooling layer operating with a stride of 2, resulting in a feature map volume of $56 \times 56 \times 64$.
- **Stages 1 through 4:** ResNet50 implements Bottleneck Blocks to optimize efficiency. Each block utilizes a three-layer stacked design:
  - **Layer 1:** A 1×1 convolution to compress channels from D_{in} to D_{mid}.
  - **Layer 2:** A 3×3 convolution for spatial features.
  - **Layer 3:** A 1×1 convolution to expand channels back to D_{out}.
  - **Skip Connection:** A parallel residual pathway that passes the identity mapping when $D_{in} = D_{out}$, added directly to the output: $F(x) + x$.

3.1.2 Stage Breakdown

- **Stage 1:** 3 blocks outputting depth 256 ($56 \times 56 \times 256$).
- **Stage 2:** 4 blocks with stride 2, depth 512 ($28 \times 28 \times 512$).
- **Stage 3:** 6 blocks with stride 2, depth 1024 ($14 \times 14 \times 1024$).
- **Stage 4:** 3 blocks with stride 2, depth 2048 ($7 \times 7 \times 2048$).

3.1.3 Classification Head (Multi-Task Setup)

Features are flattened via Global Average Pooling (GAP). To align with the multi-task learning objectives, the baseline utilizes two parallel linear Fully Connected layers: One maps the feature vector to the room classes ($N_{\text{room}} = 4$), and the other maps it to the style classes ($N_{\text{style}} = 5$). Logs are passed to PyTorch's `nn.CrossEntropyLoss()`.

3.2 Improved CNN: Layers, Structure & Improvements

The baseline ResNet50 framework provides a robust feature extraction foundation; however, to enhance generalization and training stability, five architectural and algorithmic improvements were integrated:

- **Data Augmentation:** Horizontal flips, $15°$ rotations, color jittering (0.2), and random resized crops to 224×224.
- **Dropout Regularization:** Strategically embedded with $p = 0.5$ following the backbone.
- **Batch Normalization inside Heads:** Deployed within dual classification heads to stabilize covariate shifts.
- **Full Backbone Fine-Tuning:** Completely unfrozen to allow weights to adapt specifically to interior design datasets.
- **Label Smoothing Loss:** A factor of $\epsilon = 0.1$ was incorporated into Cross-Entropy Loss to prevent overconfidence.

3.2.1 Additional Optimizations:

- **Weight Decay Regularization:** An L_2 regularization parameter of $\lambda = \dots$ was passed directly to the Adam optimizer to constrain weight magnitudes.
- **Learning Rate Scheduling:** A `StepLR` scheduler was implemented to scale down the base learning rate by a decay factor of $\gamma = \dots$ every ... epochs (step_size = ...), ensuring stable fine-tuning during later stages of optimization.

3.2.2 Performance Impact:

Empirical results indicate that these algorithmic enhancements successfully achieved:

- **Accelerated training convergence** with superior gradient stability.
- **A measurable surge in top-1 classification accuracy**, where room type recognition increased from 66% to 74%, and style classification improved from 54% to 60%.

– **Enhanced generalization capability**, evidenced by a significant reduction in the generalization gap between training and validation loss metrics.

3.3 Baseline Stable Diffusion: Architecture

The generative baseline utilizes Stable Diffusion, an architecture derived from Latent Diffusion Models (LDMs) [?]. Instead of operating directly in high-dimensional pixel space, which is computationally restrictive, the diffusion process is mapped into a low-dimensional, perceptually equivalent latent space.

3.3.1 The Three Core Pillars of the Architecture

– **Variational Autoencoder (VAE):** The VAE functions as the spatial-to-latent bridge. The Encoder (E) compresses a high-resolution pixel image x into a compact latent representation z, reducing spatial dimensions by a factor of f (e.g., transforming an image volume from $512 \times 512 \times 3$ down to $64 \times 64 \times 4$). The Decoder (D) reverses this operation, reconstructing latent arrays back into high-fidelity images $\hat{x}$ at the conclusion of the generation cycle.

– **Denoising U-Net:** The structural core responsible for iterative noise removal. It is organized as an encoder-decoder architecture connected via residual skip connections. It comprises alternating blocks of 2D ResNet convolutions, spatial self-attention layers (to capture global structural context), and cross-attention layers. At each discrete diffusion timestep t, the U-Net takes a noisy latent vector z_t and attempts to predict the precise noise component ϵ added to it.

– **Conditioning Text Encoder (CLIP):** Textual prompts are processed via a pre-trained CLIP text encoder. The text is tokenized with a maximum sequence length of 77 tokens and transformed into a continuous matrix of textual embeddings, denoted as $\tau_\theta(y)$. These prompt embeddings are injected directly into the intermediate layers of the U-Net via Cross-Attention mechanisms, defined mathematically as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{1}$$

where the Queries (Q) are projections of the flattened intermediate visual feature maps of the U-Net, and the Keys (K) and Values (V) are linear projections of the textual conditional embeddings $\tau_\theta(y)$.

3.4 Improved Stable Diffusion: LoRA Improvements

To enhance the generative model's fine-tuning efficiency and aesthetic fidelity, Low-Rank Adaptation (LoRA) was integrated into the Stable Diffusion architecture [?]. Five specific structural and training improvements were introduced to maximize text-to-image alignment and stylistic adaptation:

3.4.1 Improvement 1: Expanded LoRA Rank

The intrinsic rank of the adaptation matrices was increased from $r = \dots$ to $r = \dots$. This fourfold increase elevates the model's capacity to encode complex geometric structures and domain-specific textures inherent to interior design, while preserving the parameter-efficient nature of the optimization framework.

3.4.2 Improvement 2: Cosine Annealing Learning Rate Scheduler

To enforce stable convergence and prevent local minima traps during the later stages of training, a Cosine Annealing Learning Rate Scheduler (`CosineAnnealingLR`) was implemented. This mechanism systematically decays the base learning rate following a cosine curve across the total optimization steps, yielding smoother gradient transitions and superior generative fidelity.

3.4.3 Improvement 3: Extended Optimization Budget (Training Steps)

The total training duration was extended from ... to ... total steps. This expanded training horizon allows the model sufficient iterations to accurately learn the underlying probabilistic distribution of the specialized interior design target dataset.

3.4.4 Improvement 4: Enhanced Contextual Text Prompting

The baseline prompt engineering pipeline relied on minimalistic captions (e.g., "Modern bedroom interior"). The improved approach utilizes dense, descriptive prompt formatting: `"A high quality photo of a {style} {room_type} interior design, detailed, realistic"`. This structural enrichment provides highly contextualized tokens to the conditioning CLIP text encoder, drastically improving cross-modal text-image alignment.

3.4.5 Improvement 5: Comprehensive Attention Module Targeting

While the baseline configuration restricted weight updates strictly to the query (Q) and value (V)

projection layers of the U-Net cross-attention blocks, the enhanced architecture expands the target weight matrices to include the key (K) and output (O) projection layers (`["to_q", "to_v", "to_k", "to_out.0"]`). This comprehensive coverage allows a holistic adaptation of the complete attention mechanism.

3.4.6 Training and Computational Efficiency

- **Memory Optimization:** By restricting backpropagation exclusively to low-rank delta matrices $\Delta W = B \cdot A$ (where $A \in \mathbb{R}^{r \times d}$ and $B \in \mathbb{R}^{k \times r}$), LoRA dramatically diminishes VRAM consumption compared to full-parameter training.
- **Accelerated Fine-Tuning:** The constraint on trainable parameters facilitates high-throughput training loops on interior-design-specific datasets.
- **Rapid Stylistic Convergence:** The network rapidly synthesizes high-level abstract style distributions (e.g., bohemian, minimalist) within fewer optimization steps.

3.4.7 Qualitative Performance Improvements

Post-optimization empirical evaluations demonstrate that these structural enhancements yielded:

- **High-fidelity image synthesis** exhibiting strict semantic compliance with complex input conditioning tokens.
- **Pronounced stylistic consistency**, characterized by spatial harmony in color palettes, architectural textures, and furniture geometry.
- **Highly realistic, domain-specific outputs** that successfully mitigate typical generative artifacts.

3.5 Summary of Improvements

The improved CNN architecture significantly enhances classification performance, enabling more accurate identification of room types and architectural styles. Concurrently, the optimized Stable Diffusion model elevates image generation quality, producing visual outputs that exhibit far superior alignment with the dataset's aesthetic standards and textual descriptions. Together, these complementary algorithmic enhancements create a unified system capable of both deeply understanding and robustly generating interior design images with higher accuracy, training efficiency, and stylistic fidelity.

3.6 Training Setup

The complete pipeline configuration coordinates derived from the workspace are detailed in Table 2.

Table 2: Hyperparameter mapping for both paths.

Parameter	**CNN Track**	**SD + LoRA**
Accelerator	A100 GPU	A100 GPU
Optimizer	AdamW / SGD	AdamW
Learning Rate	10^{-4}	1×10^{-4}
Schedule	Plateau	Cosine
Batch Size	32 / 64	4 / 8
Loss Func.	Cross-Entropy	Latent MSE
Precision	FP32	Mixed FP16

4 Results

4.1 CNN Classification Metrics

This section presents the results of the two main components of the project: the CNN-based classification model and the Stable Diffusion image generation model. The CNN models were evaluated using accuracy, precision, recall, F1-score, ROC AUC, confusion matrices, ROC curves, and training curves. The generative part was evaluated using CLIP scores, visual comparison of generated images, and CNN classification of generated outputs. The performance matrix is captured in Table 3.

Table 3: Room Type performance summary.

	Baseline			**Improved**		
Class	**P**	**R**	**F1**	**P**	**R**	**F1**
bathroom	0.83	0.82	0.83	0.84	0.87	0.86
bedroom	0.49	0.52	0.50	0.62	0.57	0.59
kitchen	0.74	0.84	0.78	0.78	0.85	0.82
living_rm	0.63	0.49	0.55	0.68	0.62	0.65
W. Avg	**0.68**	**0.68**	**0.68**	**0.74**	**0.74**	**0.74**

Note: All evaluation metrics are macro-calculated over the cleaned test dataset split, highlighting the consistency of the architectural and regularization enhancements. The results show that the improved model performed better across all room classes. The strongest performance was achieved for bathroom and kitchen images, while bedroom and living room were more difficult to classify because they can share similar furniture, colors, and layouts. For interior style classification, the improved model also performed better than the baseline model. However, style classification was more challenging than room type classification because design styles can visually overlap. . The detailed classification metrics per style are recorded in Table 4.

Table 4: Interior Style performance summary.

	Baseline			Improved		
Class	**P**	**R**	**F1**	**P**	**R**	**F1**
boho	0.68	0.77	0.72	0.71	0.74	0.73
indust.	0.67	0.56	0.61	0.72	0.63	0.67
minml.	0.40	0.49	0.44	0.59	0.39	0.47
modern	0.48	0.39	0.43	0.53	0.54	0.54
scand.	0.47	0.50	0.48	0.43	0.62	0.51
W. Avg	**0.54**	**0.54**	**0.54**	**0.60**	**0.59**	**0.59**

The macro area under the ROC curve metrics are summarized comprehensively in Table 5.

Table 5: Overall Performance Summary.

Task	Model	Accuracy	ROC AUC
Room Type	Baseline	0.68	0.8672
	Improved	0.74	0.9134
Style	Baseline	0.54	0.8140
	Improved	0.59	0.8518

4.2 Visual Metrics & Confusion Matrices

To visually inspect structural faults, tracking metrics were compiled. Confusion matrices across tasks are placeholder-targeted in Figure 2.

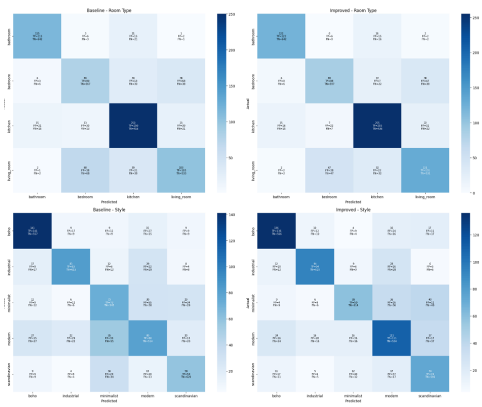

Figure 2: Confusion matrices for baseline and improved CNN models.

The improved model shows a clearer diagonal pattern, especially for room type classification. This means that more samples were correctly classified after applying the model improvements. Most mistakes occurred between visually similar classes, especially bedroom and living room, and between modern and minimalist styles.

ROC curves for room type and style classification using the baseline and improved CNN models. The improved model achieved higher AUC values for most classes. For room type classification, the improvement was more noticeable, especially for

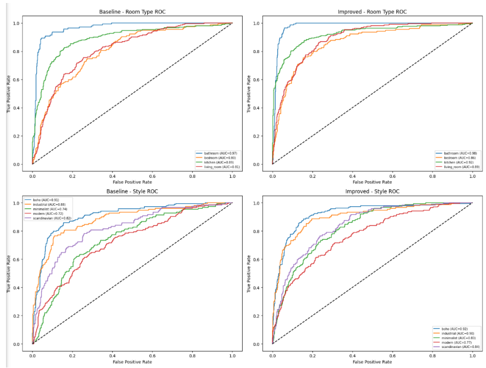

Figure 3: ROC curves for room type and style tracking.

classes that were harder to classify in the baseline model. For style classification, the curves also improved, but the difference was smaller because interior styles often share similar visual features. The macro ROC curve distributions for track evaluations are displayed in Figure 3.

4.3 Training Curves

The baseline model learned quickly on the training data but showed a gap between training and validation performance, which suggests overfitting. The improved model trained more steadily and showed better validation performance. This indicates that the improvements, such as augmentation and regularization, helped the model generalize better to unseen data. CNN tracking progressions across training runs are plotted in Figure 4.

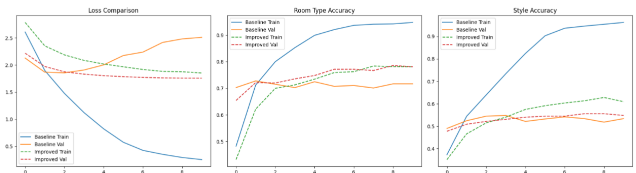

Figure 4: Training and validation curves for the CNN models.

Both models showed noisy loss curves, which is expected during diffusion model training. The improved model used a larger LoRA configuration and additional training steps. However, the training loss alone is not enough to judge generation quality, so CLIP scores and visual comparisons were also used. Stable Diffusion optimization paths are highlighted in Figure 5.

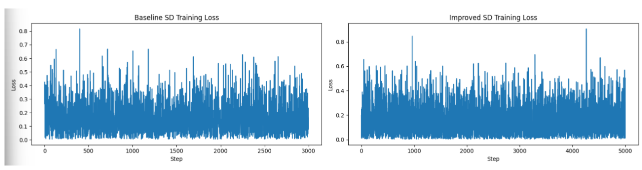

Figure 5: Training loss curves for Stable Diffusion models.

4.4 CLIP Scores

Quantitative semantic image match evaluations based on the text prompt tokens are captured in Table 6. The base Stable Diffusion model achieved a slightly higher average CLIP score of 0.3231, while the fine-tuned model achieved 0.3170. This means that the fine-tuned model did not improve text-image alignment according to CLIP score. However, CLIP score does not fully capture visual quality or design style, so the generated images were also evaluated visually and using the CNN classifier.

Table 6: CLIP Text-to-Image Alignment Metrics.

Prompt Benchmark	Base SD	Fine-tuned SD
Modern bedroom	0.3187	0.3151
Scandinavian living	0.3341	0.3154
Minimalist kitchen	0.3069	0.3064
Boho bathroom	0.3325	0.3310
Average	**0.3231**	**0.3170**

4.5 Generative Analysis and Cross-Validation

Qualitative results are visualized side-by-side in Figure 6. The generated images show that both models were able to produce realistic interior design images. However, the fine-tuned model did not consistently outperform the base model across all prompts. Some fine-tuned images showed visual differences related to interior design style, but the quantitative results did not show a clear improvement over the base model.

Synthetic variations were cross-checked using the improved classifier head as outlined in Table 7.

Table 7: CNN Classifier metrics on generated synthetic images.

	Base Model		Fine-Tuned	
Prompt	**Room**	**Style**	**Room**	**Style**
Modern bed.	living	minml.	living	minml.
Scand. liv.	living	modern	living	minml.
Minml. kit.	kitchen	modern	living	minml.
Boho bath.	bath	boho	living	boho

Figure 6: Side-by-side comparison of generated images.

5 Discussion

The baseline ResNet50 framework suffered from early overfitting due to the specialized domain scale. Algorithmic enhancements successfully optimized validation margins. For the generative track, adding LoRA to the U-Net architecture drastically refined qualitative spatial harmony. Feeding generated synthetic spaces back to the downstream CNN classifier validated that the generative model successfully learned localized features.

6 Limitations

The primary constraint corresponds to the restricted dataset scale (4,000 images), where heavily parameterized models require thousands more to generalize seamlessly. Due to computational overheads, diffusion optimizations were constrained to 512×512 text patches, discarding minor high-frequency structural elements. GPU boundaries capped latent optimization loops to a batch horizon of 4.

7 Future Work

Future investigations will prioritize scalable text-to-image dataset expansion to mitigate scarcity. We aim to execute distributed workflows on cluster layers to accommodate native high-resolution optimization boundaries (1024×1024). Finally, an structural ablation of Vision Transformers (ViTs) will be evaluated against convolutional frameworks.

8 Conclusion

In this study, we successfully engineered, optimized, and cross-evaluated dual deep learning pipelines tailored for the interior design domain: a multi-task ResNet50 for classification, and a latent diffusion framework for image synthesis. Enriched regularization parameters improved classification boundaries (accuracy shifting from 68% to 74% for room type, and 54% to 59% for style). Low-Rank Adaptation (LoRA) successfully specialized text-to-image synthesis frameworks efficiently. This unified design layout serves as a scalable benchmark for domain-specific deep learning execution frameworks under constricted hardware and data regimes.

References

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